

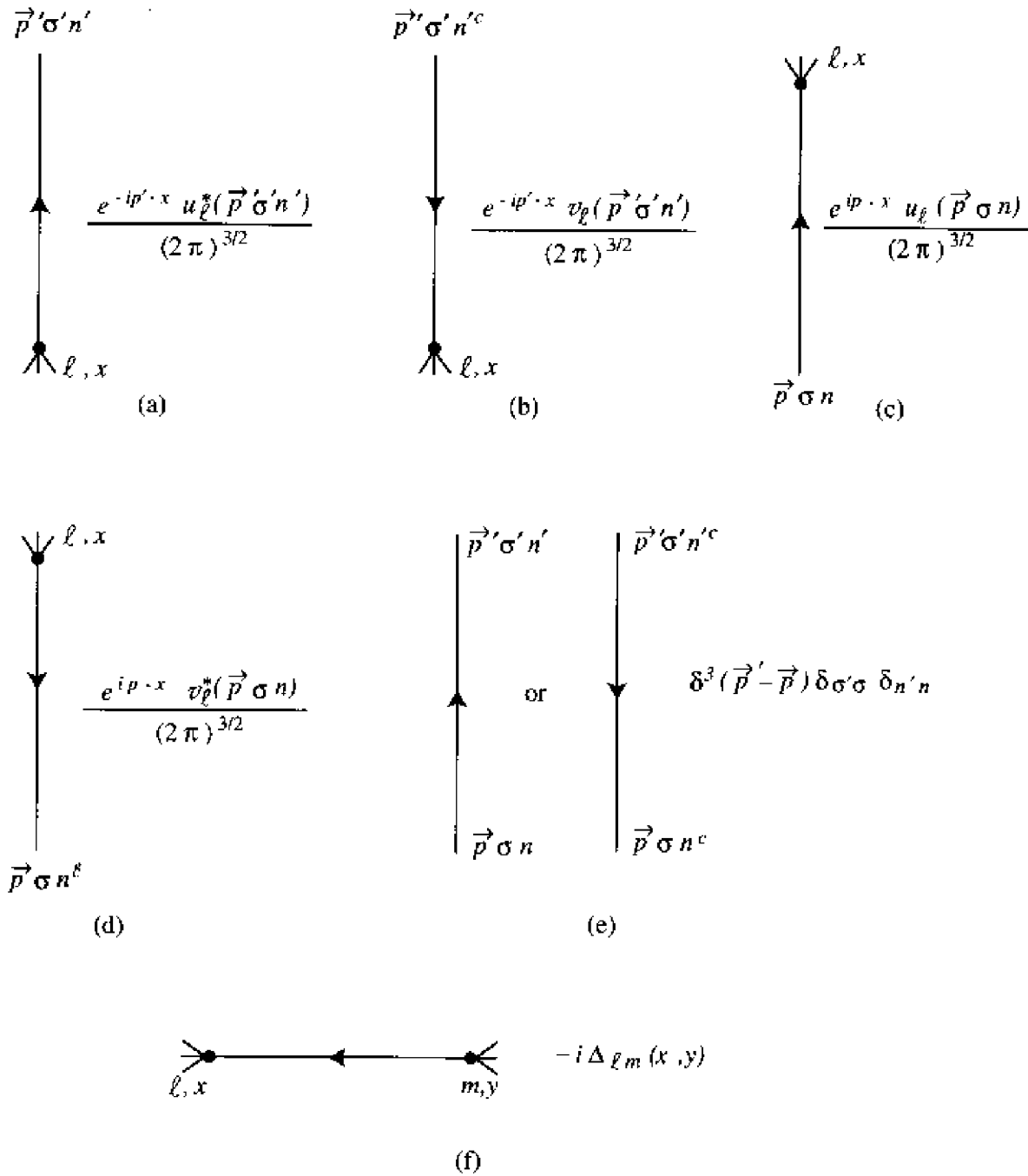


Figure 6.1. Graphical representation of pairings of operators arising in the coordinate-space evaluation of the S -matrix. The expressions on the right are the factors that must be included in the coordinate-space integrand of the S -matrix for each line of the Feynman diagram.

The rules for calculating the S -matrix are conveniently summarized in terms of *Feynman diagrams*. (See Figure 6.1.) The diagrams consist of points called *vertices*, each representing one of the $\mathcal{H}_i(x)$, and *lines*, each representing the pairing of a creation with an annihilation operator. More specifically:

(a) The pairing of a final particle with a field adjoint in one of the $\mathcal{H}(x)$